

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2020

BIOLOGY PAPER 1

8:30 am – 11:00 am (2 hours 30 minutes)

This paper must be answered in English

GENERAL INSTRUCTIONS

- (1) There are **TWO** sections, A and B, in this Paper. You are advised to finish Section A in about 35 minutes.
- (2) Section A consists of multiple-choice questions in this question paper. Section B contains conventional questions printed separately in Question-Answer Book **B**.
- (3) Answers to Section A should be marked on the Multiple-choice Answer Sheet while answers to Section B should be written in the spaces provided in Question-Answer Book **B**. **The Answer Sheet for Section A and the Question-Answer Book B for Section B will be collected separately at the end of the examination.**

INSTRUCTIONS FOR SECTION A (MULTIPLE-CHOICE QUESTIONS)

- (1) Read carefully the instructions on the Answer Sheet. After the announcement of the start of the examination, you should first stick a barcode label and insert the information required in the spaces provided. No extra time will be given for sticking on the barcode label after the 'Time is up' announcement.
- (2) When told to open this book, you should check that all the questions are there. Look for the words '**END OF SECTION A**' after the last question.
- (3) All questions carry equal marks.
- (4) **ANSWER ALL QUESTIONS.** You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber. You must mark the answers clearly; otherwise you will lose marks if the answers cannot be captured.
- (5) You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.
- (6) No marks will be deducted for wrong answers.

Not to be taken away before the end of the examination session

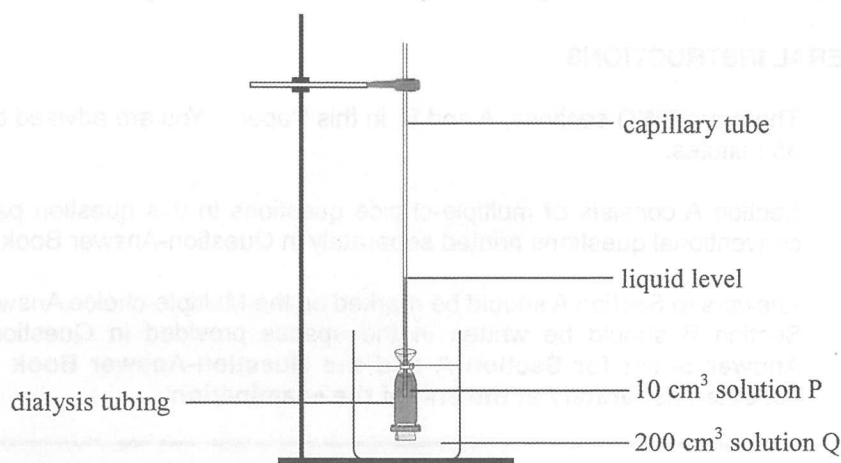
There are 36 questions in this section.

The diagrams in this section are NOT necessarily drawn to scale.

1. Which of the following sub-cellular structures can be found in a prokaryotic cell?

- A. cell wall, ribosome, cell membrane
- B. ribosome, cell membrane, chloroplast
- C. cell wall, cell membrane, mitochondria
- D. cell membrane, chloroplast, mitochondria

Directions: Questions 2 and 3 refer to the diagram below, which shows a set-up for investigating the permeability of dialysis tubing. 10 cm³ of solution P was added to the dialysis tubing, with one end tied and the other end connected to a capillary tube. The dialysis tubing was then placed in a beaker with 200 cm³ of solution Q. When solutions P and Q are mixed, a blue colour will be observed.



2. At the end of the investigation, the liquid level inside the capillary tube had risen and only the solution inside the dialysis tubing became blue. Which of the following conclusions can be drawn from the results?

- (1) Solute of solution P can pass through the dialysis tubing.
- (2) Solute of solution Q can pass through the dialysis tubing.
- (3) There is a net movement of water into the dialysis tubing.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

3. If the investigation is repeated with water instead of solution Q, which of the following results will be obtained?

- A. The liquid level in the capillary tube will not rise at all.
- B. The liquid level in the capillary tube will rise to a lower level.
- C. The liquid level in the capillary tube will rise to a higher level.
- D. The liquid level in the capillary tube will rise to the same level.

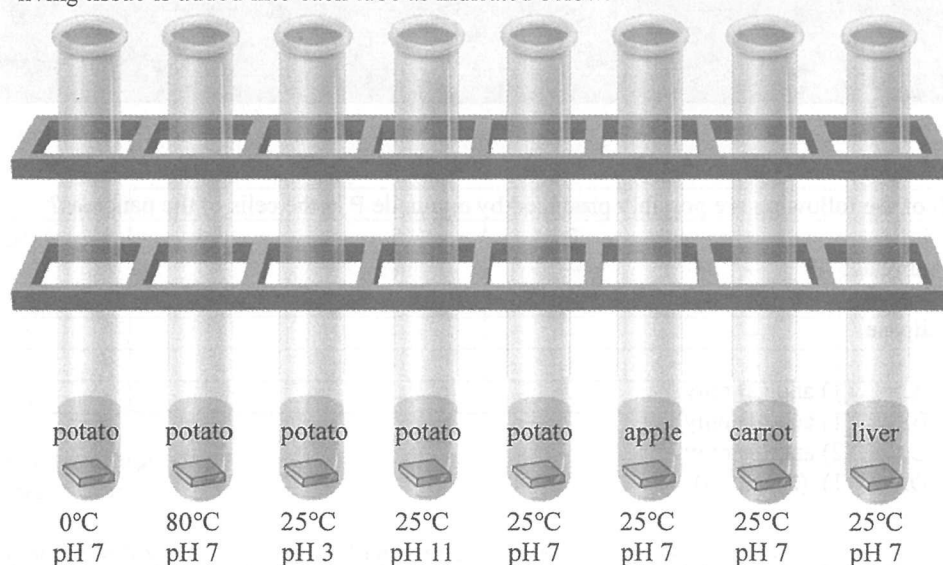
4. Which of the following descriptions of anaerobic respiration in muscles is correct?

- A. It involves glycolysis.
- B. It produces lactic acid and carbon dioxide.
- C. It takes place in the matrix of mitochondria.
- D. It takes place on the inner membrane of mitochondria.

5. Which of the following combinations correctly matches the stages of cellular respiration and the metabolites produced?

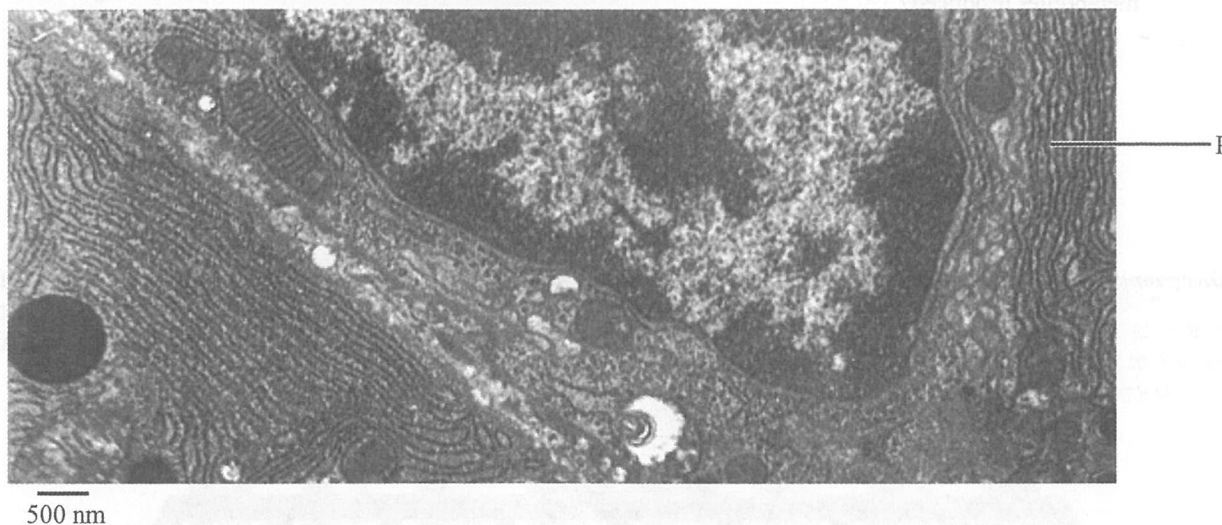
	<i>Stage</i>	<i>Metabolite produced</i>
A.	Glycolysis	acetyl-CoA
B.	Glycolysis	ATP
C.	Krebs Cycle	NAD
D.	Krebs Cycle	pyruvate

Directions: Questions 6 and 7 refer to the diagram below, which shows a set-up for investigating the activity of catalase in living tissues. Catalase is an enzyme which can break down hydrogen peroxide. Each test tube contains the same amount of hydrogen peroxide solution at the same concentration. A piece of living tissue is added into each tube as indicated below:



6. How many independent variables are being studied in this investigation?
- 2
 - 3
 - 4
 - 8
7. Which of the following control variables is most important for a fair comparison in the above investigation?
- mass of living tissues
 - shape of living tissues
 - volume of living tissues
 - surface area of living tissues
8. Which of the following descriptions of the active site of an enzyme is correct?
- (1) It can be used again.
 - (2) It is the part of the enzyme on which its substrate can fit.
 - (3) Its shape is determined by the amino acid sequence of the enzyme.
- (1) and (2) only
 - (1) and (3) only
 - (2) and (3) only
 - (1), (2) and (3)

9. The electron micrograph below shows an organelle P:



- Which of the following are possibly produced by organelle P in the cells of the pancreas?
- (1) amylase
 - (2) insulin
 - (3) lipase
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)
10. Which of the following explain why a person cannot swallow food and talk at the same time?
- (1) The epiglottis covers the opening of the trachea during swallowing.
 - (2) Swallowing is a reflex action while speaking is a voluntary action.
 - (3) Air must flow through the larynx to produce sounds.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)
11. Which of the following descriptions of breathing is correct?
- A. The air rushes in to expand the lungs.
 - B. The trachea expands to draw in more air.
 - C. Pressure in the lungs decreases to draw in air.
 - D. The expansion of the lungs pushes the diaphragm downward.
12. Which of the following descriptions of the hepatic portal vein is correct?
- A. It carries blood away from the liver.
 - B. It carries blood with lower oxygen content than that in the hepatic vein.
 - C. It carries blood with lower amino acid content than that in the hepatic artery after meals.
 - D. It carries blood with lower glucose content than that in the hepatic vein when one is hungry.

13. Both the villi of the small intestine and the air sacs of the lung are sites for material exchange. Which of the following is an adaptive feature common to the epithelium of both structures?

- A. The epithelium is one cell thick.
- B. There is a water film on the surface of the epithelium.
- C. The epithelial cells contain a large number of mitochondria.
- D. The epithelial cells have a modified cell membrane to increase surface area.

14. Which of the following combinations correctly matches the water movement in the human body with its major driving force?

- | | <i>Water movement</i> | <i>Major driving force</i> |
|----|--|----------------------------|
| A. | water in tissue fluid enters lymph vessels | osmosis |
| B. | water in blood leaves capillaries at the arterial end | hydrostatic pressure |
| C. | water in tissue fluid enters capillaries at the venous end | active transport |
| D. | water enters capillaries from the lumen of the small intestine | diffusion |

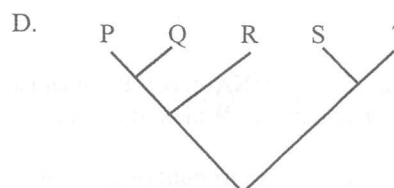
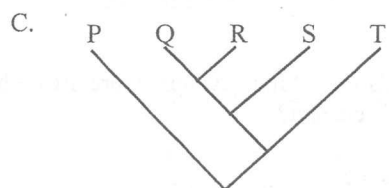
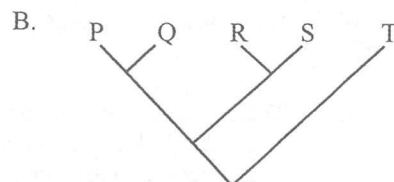
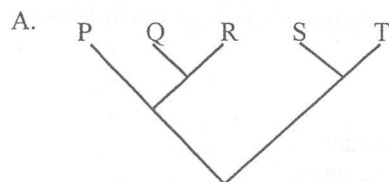
15. The table below shows the presence or absence of some traits in five species:

Species	Trait				
	1	2	3	4	5
P	+	+	—	+	—
Q	+	+	—	—	—
R	+	—	—	—	+
S	+	—	+	—	+
T	—	—	—	—	—

Key:

- + presence of trait
- absence of trait

Which of the following evolutionary trees best illustrates the phylogenetic relationship of the five species?



16. Australia has some plant and animal species which cannot be found elsewhere in the world. Which of the following is likely to be the key factor that led to the occurrence of these unique plant and animal species in Australia?

- A. isolation
- B. mutation
- C. competition
- D. environmental stress

17. 'Transgenic organisms produced by recombinant DNA technology have more potential in terms of evolution than those produced by traditional breeding.'

Which of the following is the best reason for this?

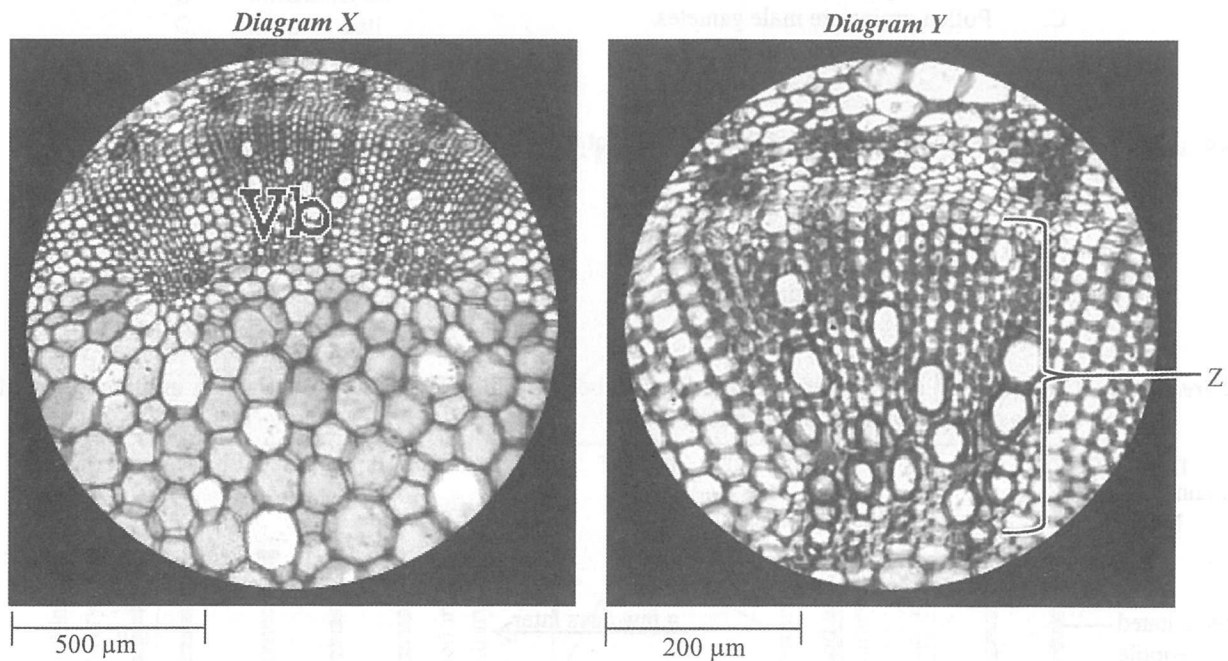
- A. Recombinant DNA technology creates new species.
- B. Recombinant DNA technology produces new phenotypes.
- C. Recombinant DNA technology transfers genes within a species.
- D. Recombinant DNA technology transfers genes between different species.

Directions: Questions 18 and 19 refer to the following passage about the discovery of DNA structure:

Many scientists tried to uncover the structure of DNA. In 1952, Franklin took the first X-ray photograph of DNA, which revealed its helical shape. Her colleague, Wilkins, showed some of Franklin's unpublished findings to Watson without her knowledge. Shortly after, Watson and Crick made a crucial advance when they proposed that the DNA molecule was made up of two chains of nucleotides paired to form a double helix. In 1962, Watson, Crick and Wilkins were awarded the Nobel Prize.

18. Which of the following aspects of the Nature of Science can be exemplified in the above story?
- (1) A good scientific experiment must include carefully designed controls.
 - (2) Scientists are both collaborative and competitive by nature.
 - (3) Doing science requires imagination and creativity.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)
19. Which of the following descriptions of the molecular structure of DNA proposed by Watson and Crick is correct?
- A. The two chains run in opposite directions.
 - B. The bases link up the nucleotides to form a chain.
 - C. The sugar that makes up nucleotides has two types.
 - D. The phosphate that makes up nucleotides has at least four types.
20. The amount of DNA in cell P immediately before mitosis is x . After division, there are 4 chromosomes in each daughter cell. Which of the following descriptions is correct?
- A. The amount of DNA in the daughter cell is $0.5x$.
 - B. The amount of DNA in each chromosome is $0.25x$.
 - C. There are 8 chromosomes in the diploid state of cell P.
 - D. There are 8 chromosomes in cell P immediately before division.
21. In flowering plants, asexual reproduction is considered less favourable than sexual reproduction in terms of natural selection because offspring produced from asexual reproduction
- A. are genetically identical to each other.
 - B. have keen competition with each other.
 - C. cannot invade and colonise new environments.
 - D. can grow rapidly only in favourable conditions.

Directions: Questions 22 to 24 refer to the photomicrographs below, which show the stem section of a plant observed under a microscope. Diagram Y shows a higher magnification of the vascular bundle (Vb) in Diagram X:



22. Which of the following combinations shows the correct steps for using the microscope in order to obtain the image in Diagram Y from that of Diagram X?

	<i>Step 1</i>	<i>Step 2</i>
A.	Move the slide towards the observer so that Vb is in the centre of the field of view.	Change the objective from 4X to 10X and adjust the focus.
B.	Change the objective from 10X to 40X and adjust the focus.	Move the slide towards the observer so that Vb is in the centre of the field of view.
C.	Move the slide away from the observer so that Vb is in the centre of the field of view.	Change the objective from 4X to 10X and adjust the focus.
D.	Change the objective from 10X to 40X and adjust the focus.	Move the slide away from the observer so that Vb is in the centre of the field of view.

23. Which of the following descriptions of the function of tissue Z shown in Diagram Y is correct?

- A. It transports proteins upwards.
- B. It transports minerals upwards.
- C. It transports water downwards.
- D. It transports sugars downwards.

24. Which of the following structures is commonly found in all the cells shown in Diagram Y?

- A. vacuole
- B. cell wall
- C. cytoplasm
- D. cell membrane

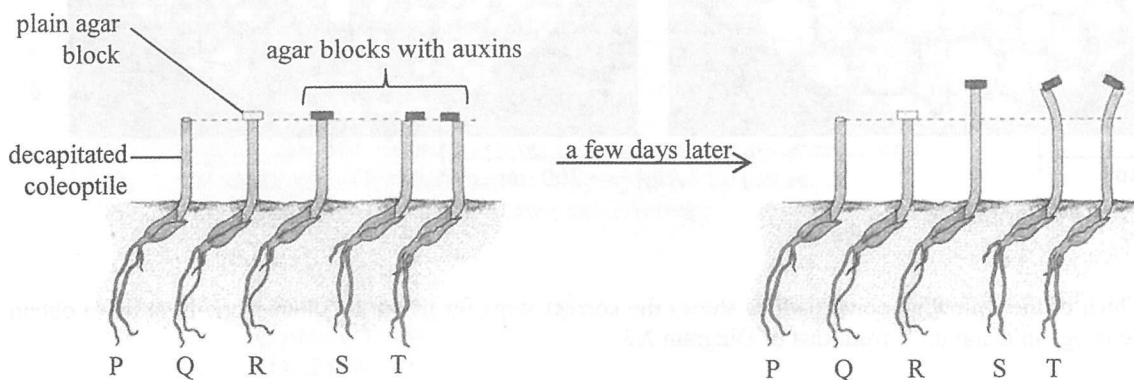
25. Which of the following descriptions of floral structures is correct?

- A. The ovule forms seeds.
- B. The ovary forms seed coats.
- C. Pollen grains are male gametes.
- D. The filament contains pollen grains.

26. On which of the following parts of the root can root hairs be found?

- A. root cap
- B. region of elongation
- C. region of cell division
- D. region of differentiation

Directions: Questions 27 and 28 refer to the diagram below, which shows an investigation into the effect of auxins on the growth of plant shoots:



27. Which of the following serve(s) as the control set-up(s) in this investigation?

- A. P only
- B. R only
- C. P and Q only
- D. Q and R only

28. Which of the following conclusions can be drawn based on the results of the investigation?

- A. The tip of the coleoptile produces auxins.
- B. Auxins stimulate the growth of coleoptiles.
- C. The coleoptile shows positive phototropism.
- D. Coleoptiles stop growing when the tip is removed.

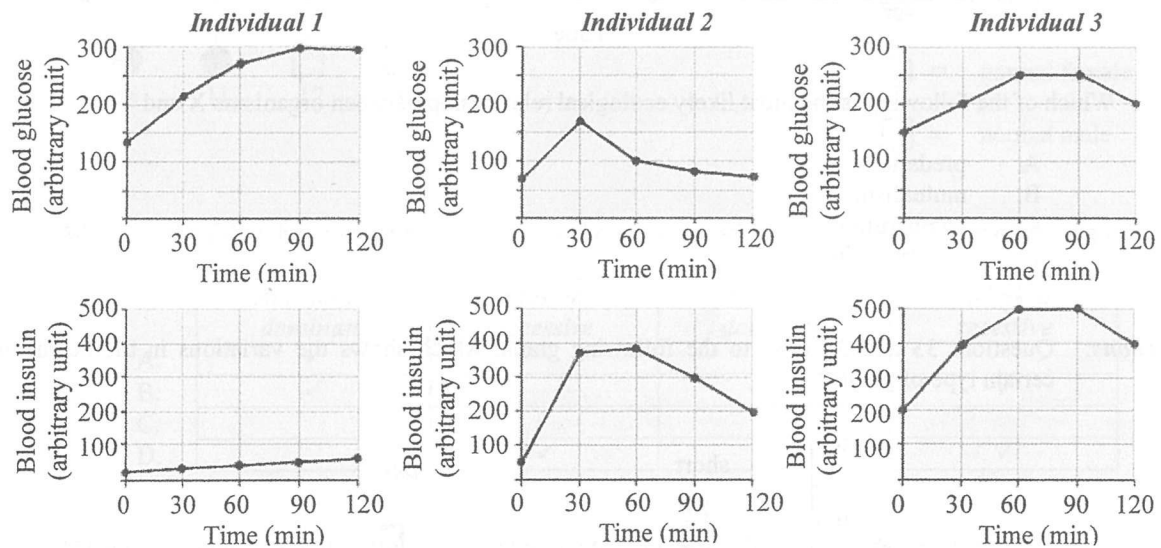
29. Which of the following combinations of causative agents and ways of transmission of infectious diseases is correct?

	<i>Infectious disease</i>	<i>Causative agent</i>	<i>Way of transmission</i>
A.	cholera	virus	food
B.	cholera	bacterium	body fluid
C.	hepatitis B	virus	body fluid
D.	hepatitis B	bacterium	food

30. Which of the following provides immunity to the human body?

- A. phagocytosis
- B. inflammation
- C. memory cell
- D. formation of blood clots

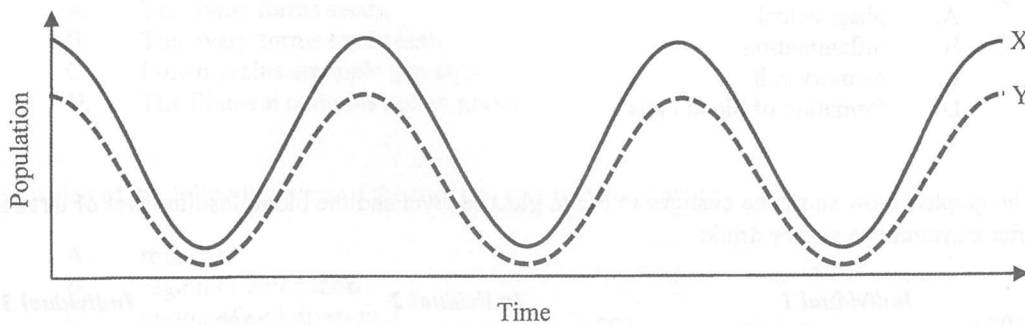
31. The graphs below show the changes in blood glucose level and the blood insulin level of three individuals after consuming a sugary drink:



Which of the following combinations correctly shows the health conditions of these three individuals?

- | | Individual 1 | Individual 2 | Individual 3 |
|----|---------------------|---------------------|---------------------|
| A. | No diabetes | Type 1 diabetes | Type 2 diabetes |
| B. | Type 1 diabetes | Type 2 diabetes | No diabetes |
| C. | Type 2 diabetes | No diabetes | Type 1 diabetes |
| D. | Type 1 diabetes | No diabetes | Type 2 diabetes |

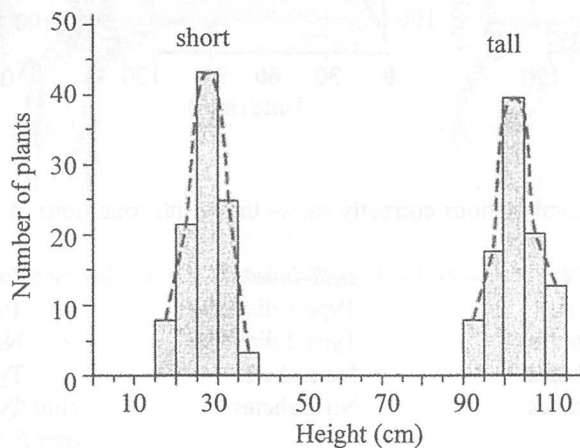
32. The following graph shows the change in the populations of organisms X and Y living in the same habitat:



Which of the following is the most likely ecological relationship between organisms X and Y?

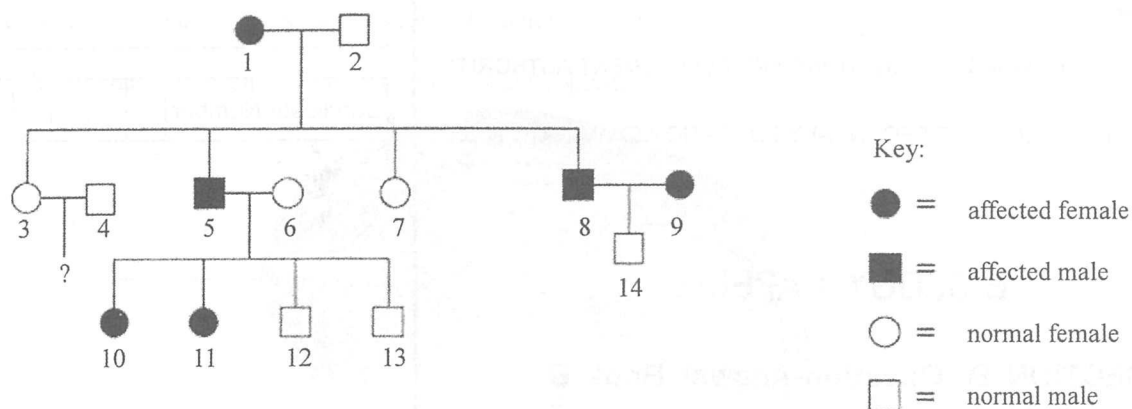
- A. predation
- B. mutualism
- C. competition
- D. commensalism

Directions: Questions 33 and 34 refer to the following graph, which shows the variations in the height of a certain type of plant:



33. Which of the following conclusions can be drawn from the above graph?
- A. The short and tall plants are of two different species.
 - B. The two traits, short and tall, are controlled by a pair of alleles.
 - C. More samples should be taken to cover the full range of heights.
 - D. The height of the plants displays the properties of continuous and discontinuous variations.
34. Which of the following factors contributes *least* to the variations shown?
- A. the height of the parental plants
 - B. the light intensity in the environment
 - C. the independent assortment of chromosomes
 - D. the oxygen concentration in the environment

Directions: Questions 35 and 36 refer to the pedigree below, which shows the inheritance of a genetic disorder.



35. Which of the following combinations correctly shows the possible types of inheritance of this disorder?

	<i>Autosomal dominant</i>	<i>Autosomal recessive</i>	<i>X-linked dominant</i>	<i>X-linked recessive</i>
A.	✓		✓	
B.	✓			✓
C.		✓	✓	
D.		✓		✓

36. What is the probability of individual 3 and individual 4 giving birth to a normal male?

- A. 0
- B. 0.25
- C. 0.5
- D. 1

END OF SECTION A

Go on to Question-Answer Book B for questions on Section B